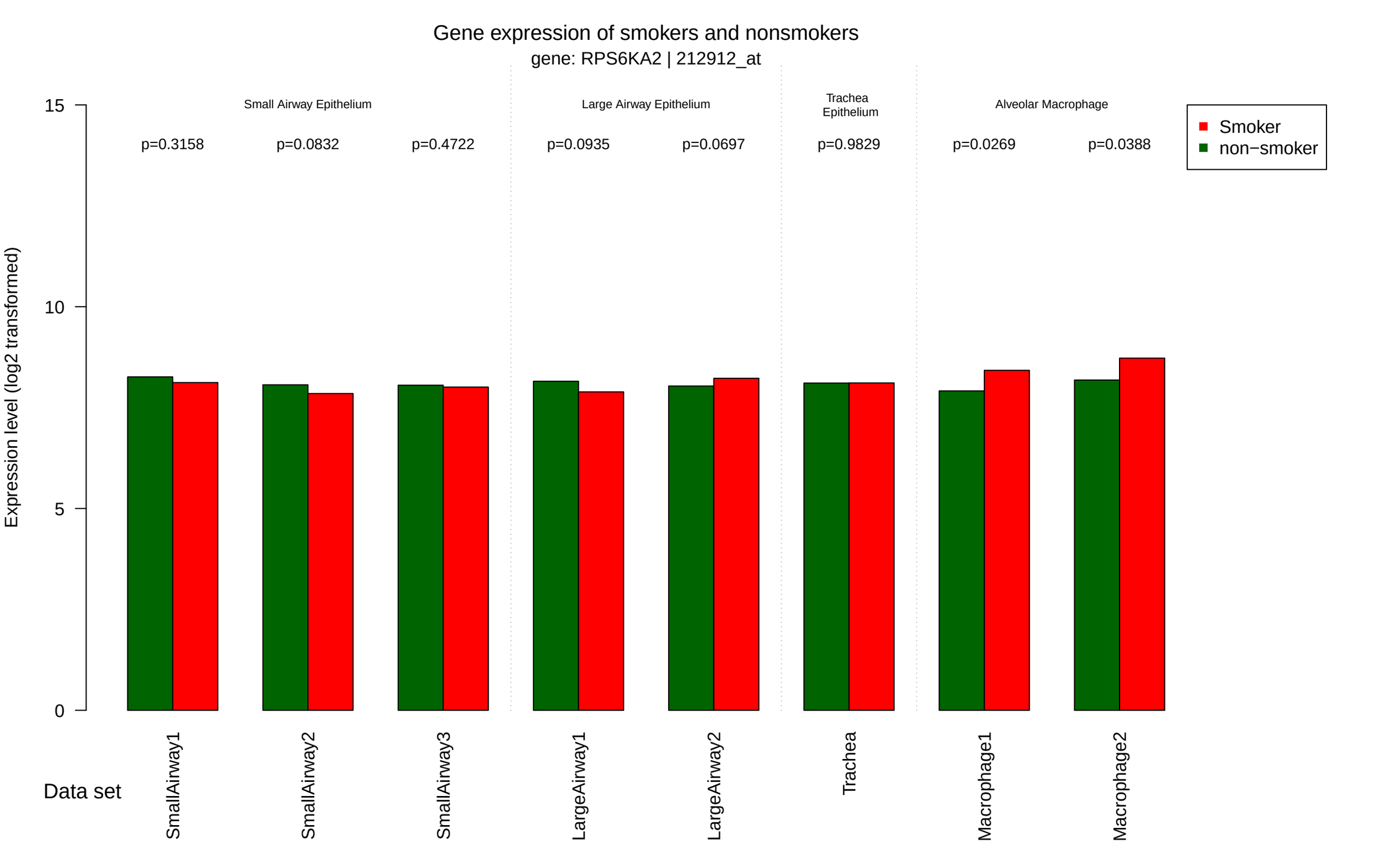
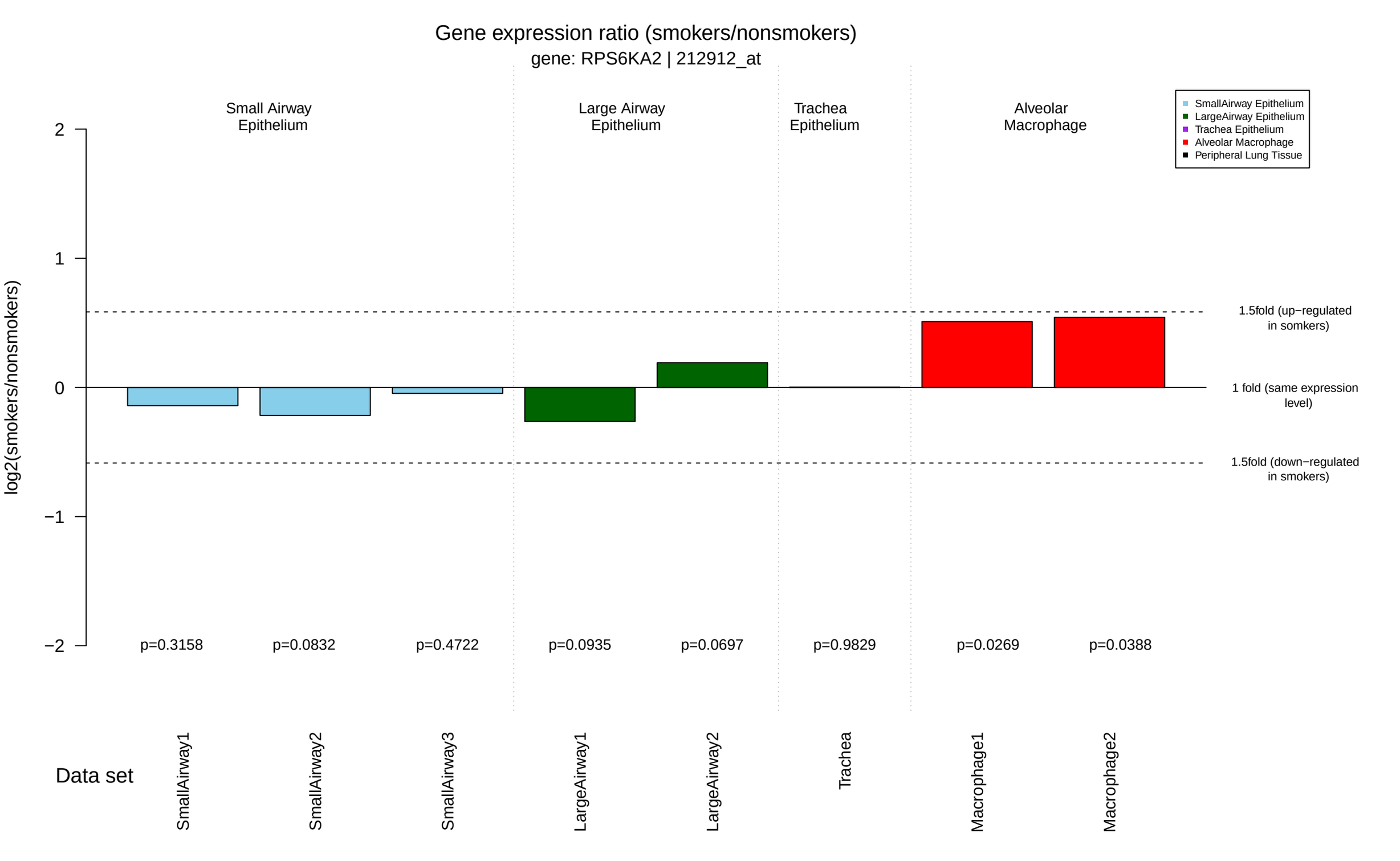
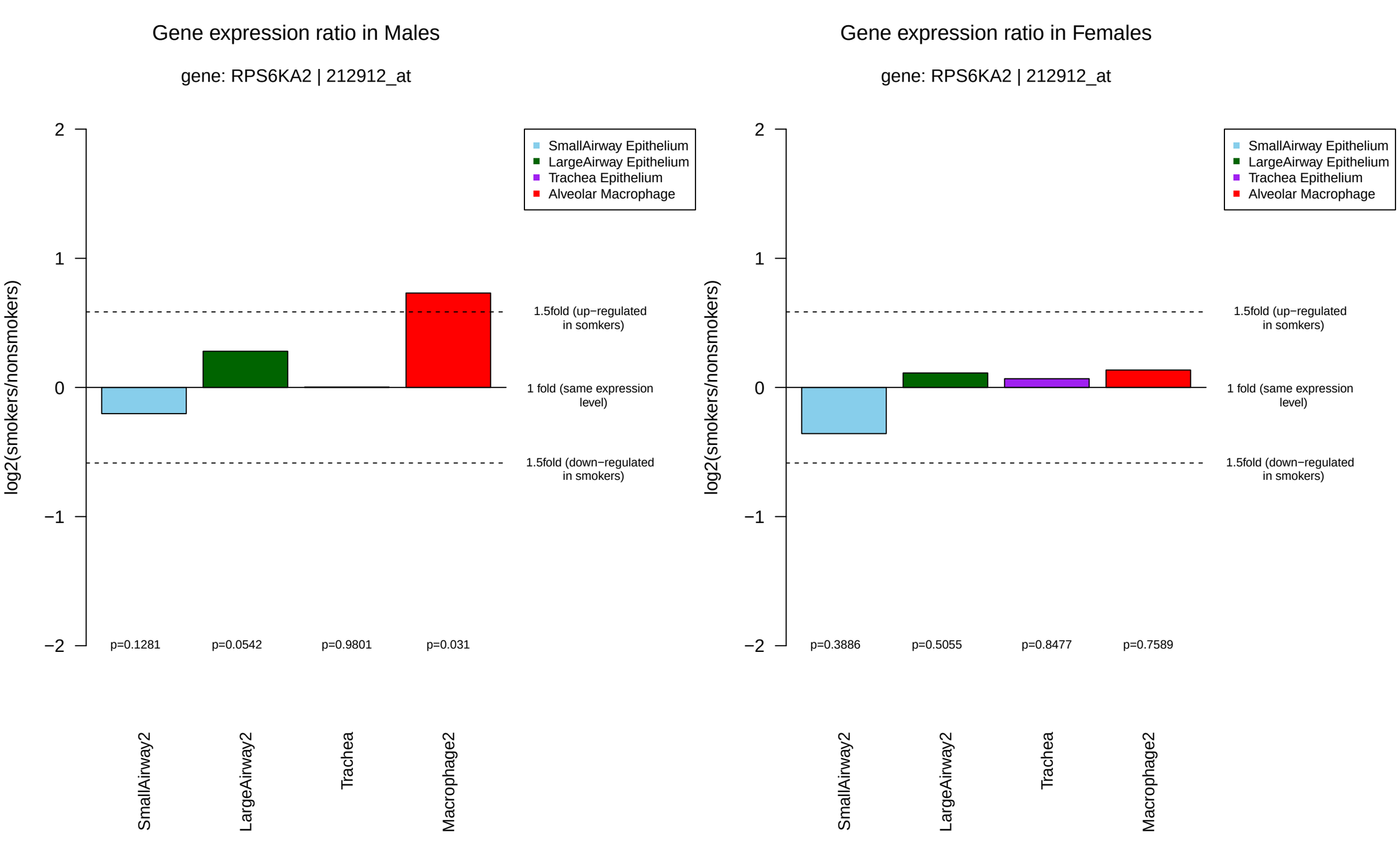
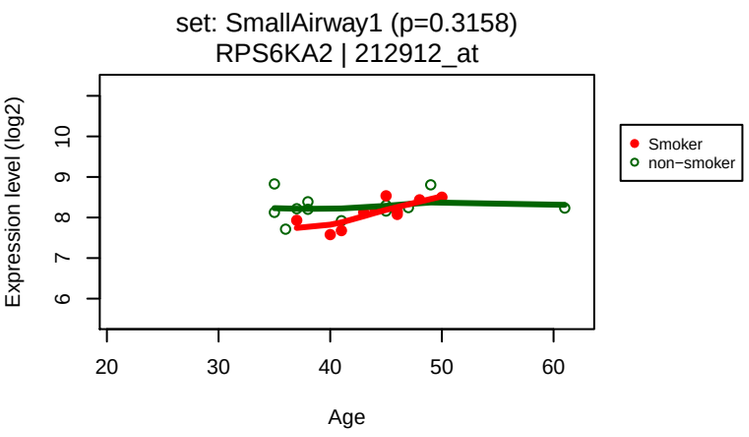
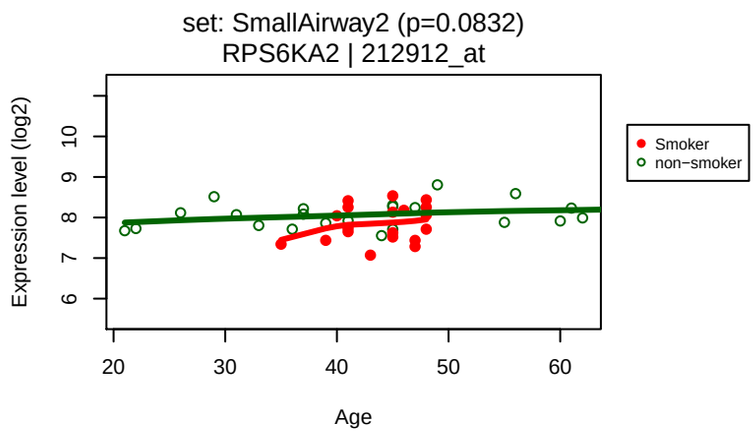
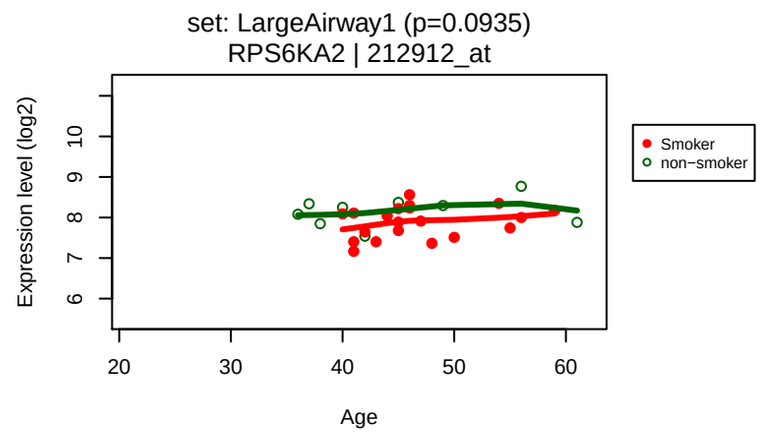

Gene expression of smokers and nonsmokers
gene: RPS6KA2 | 212912_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3158
p=0.0832
p=0.4722
p=0.0935
p=0.0697
p=0.9829
p=0.0269
p=0.0388
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

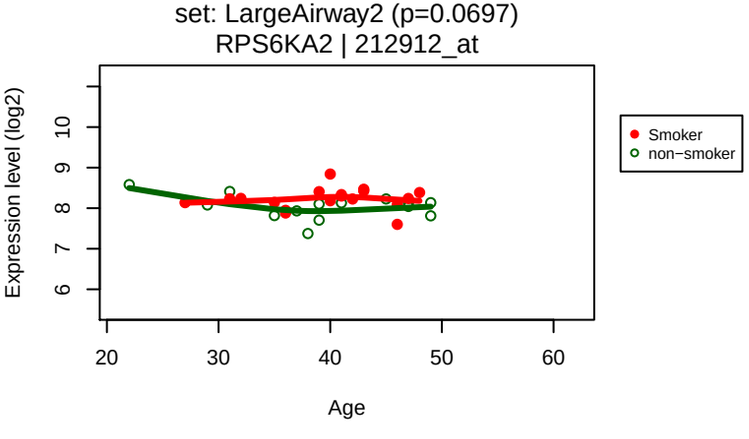

Gene expression ratio (smokers/nonsmokers)
gene: RPS6KA2 | 212912_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3158
p=0.0832
p=0.4722
p=0.0935
p=0.0697
p=0.9829
p=0.0269
p=0.0388
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

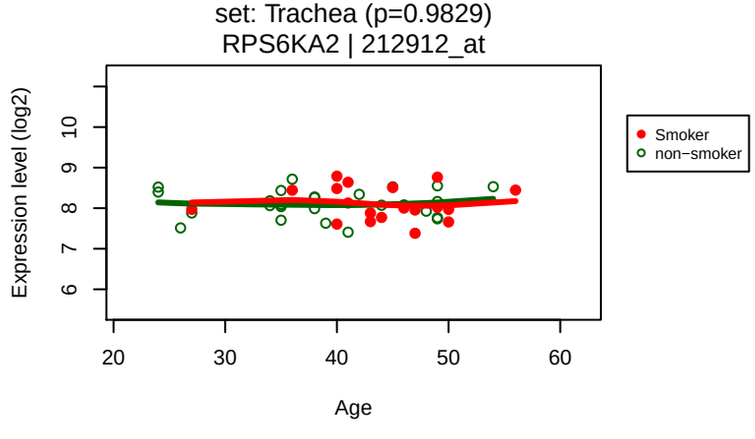

Gene expression ratio in Males
gene: RPS6KA2 | 212912_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1281
p=0.0542
p=0.9801
p=0.031
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RPS6KA2 | 212912_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3886
p=0.5055
p=0.8477
p=0.7589
SmallAirway2
LargeAirway2
Trachea
Macrophage2

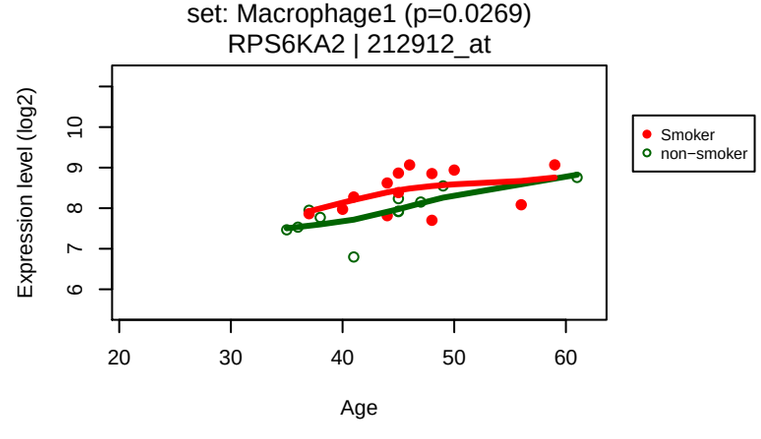

set: SmallAirway1 (p=0.3158)
RPS6KA2 | 212912_at
Expression level (log2)
Age
Smoker
non−smoker

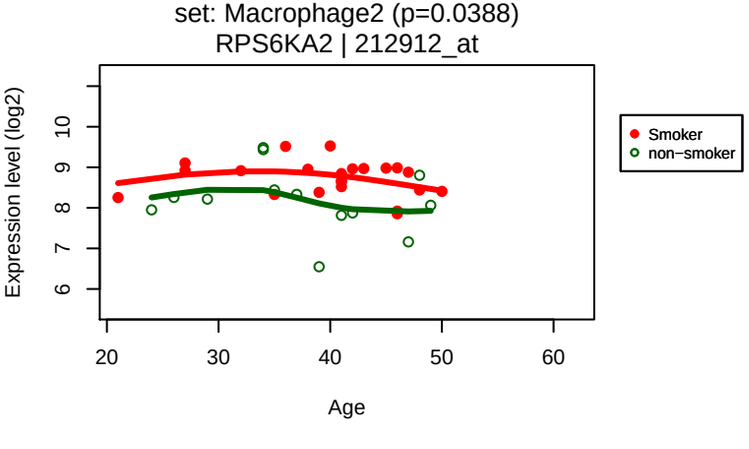

set: SmallAirway2 (p=0.0832)
RPS6KA2 | 212912_at
Expression level (log2)
Age
Smoker
non−smoker

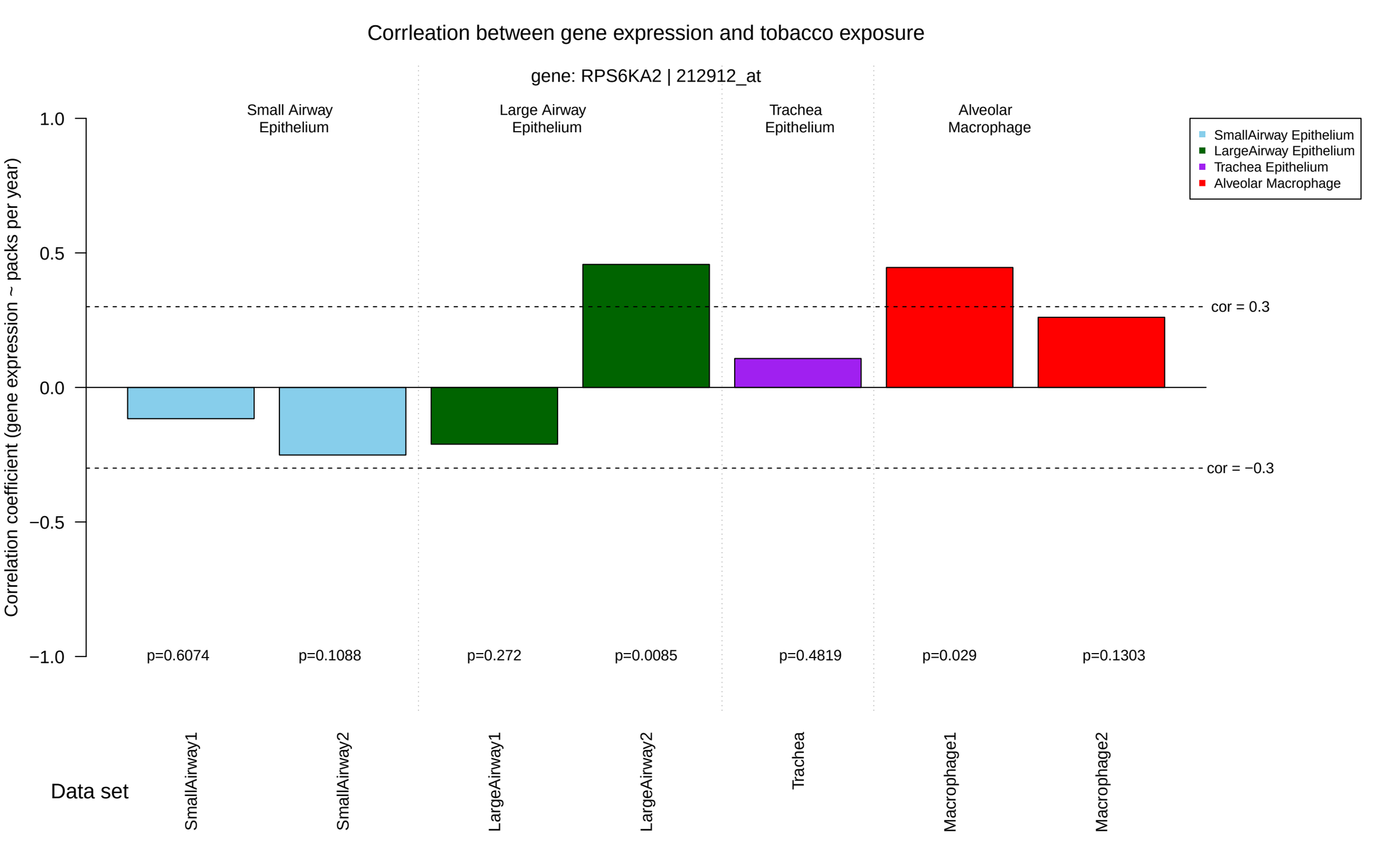

set: LargeAirway1 (p=0.0935)
RPS6KA2 | 212912_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0697)
RPS6KA2 | 212912_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.9829)
RPS6KA2 | 212912_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0269)
RPS6KA2 | 212912_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0388)
RPS6KA2 | 212912_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RPS6KA2 | 212912_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.6074
p=0.1088
p=0.272
p=0.0085
p=0.4819
p=0.029
p=0.1303
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2